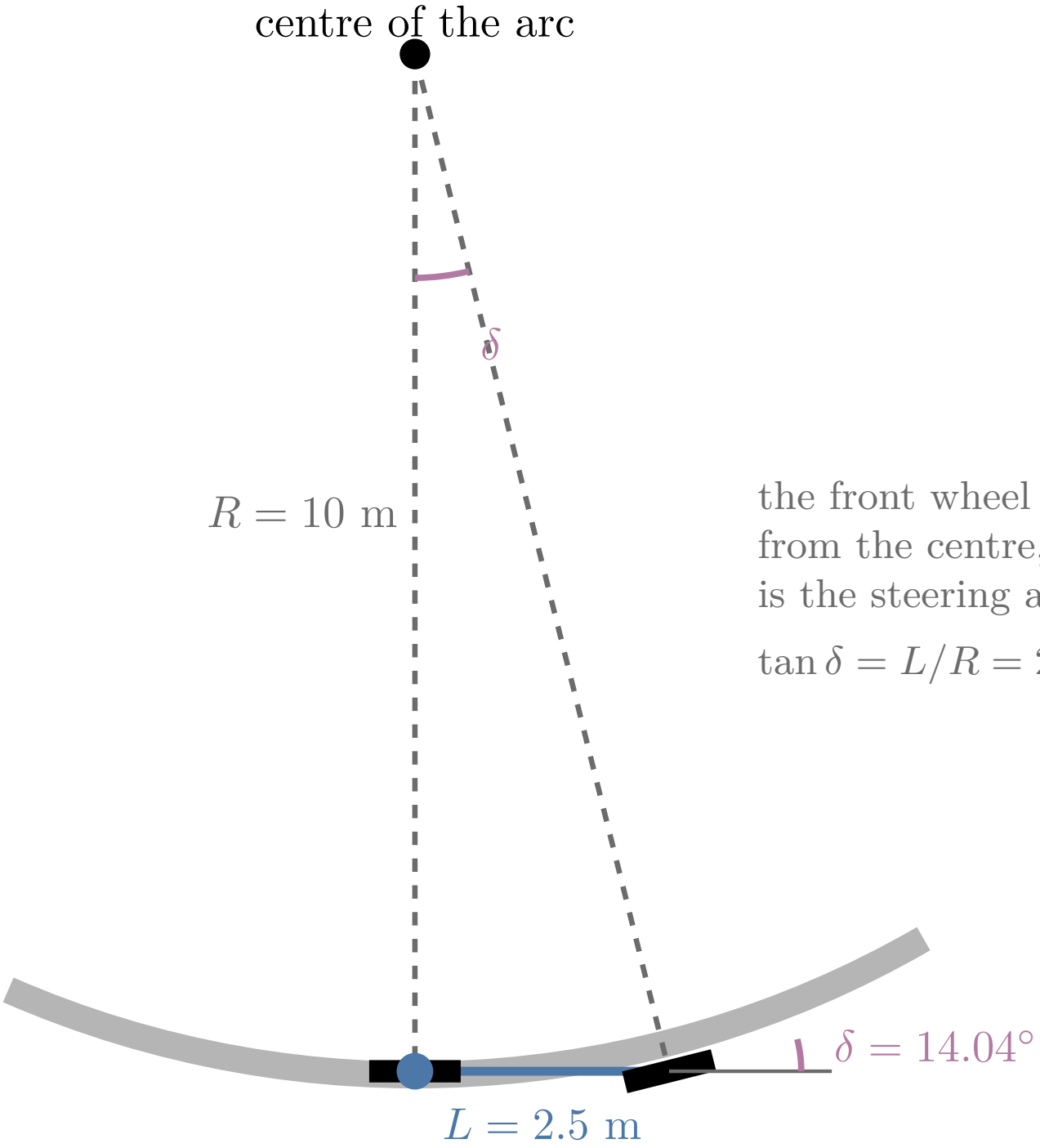


the front wheel points square to the line from the centre, so the angle at the centre is the steering angle:

$\tan \delta = L/R = 2.5/10 = 0.25$